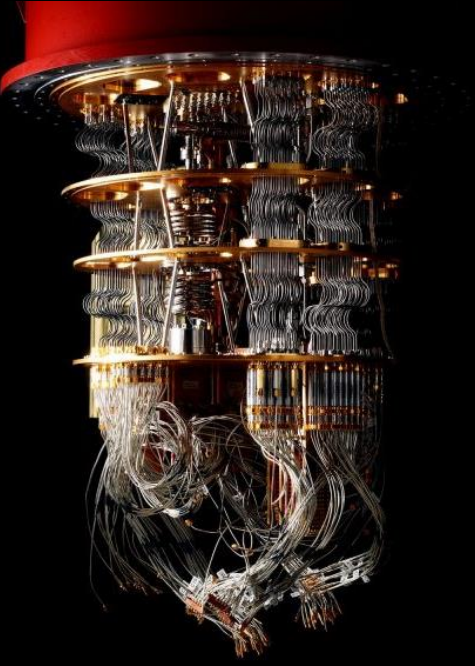




Transverse-field Ising model (TFIM)

$$H = -J \sum_{j=1}^N Z_j Z_{j+1} - h \sum_{j=1}^N X_j = H_{ZZ} + H_Z$$

$$[H_{ZZ}, H_Z] \neq 0$$

Tool #1: Jordan-Wigner transformation.

$$Z_j = \left(\prod_{k=1}^{j-1} X_k \right) (c_j^\dagger + c_j)$$

$$Z_{j+1} = \left(\prod_{k=1}^j X_k \right) (c_{j+1}^\dagger + c_{j+1}) = \left(\prod_{k=1}^{j-1} X_k \right) X_j (c_{j+1}^\dagger + c_{j+1})$$

$$Z_j Z_{j+1} = \left(\prod_{k=1}^{j-1} X_k \right) (c_j^\dagger + c_j) \left(\prod_{k=1}^{j-1} X_k \right) X_j (c_{j+1}^\dagger + c_{j+1})$$

$$Z_j Z_{j+1} = \left(\prod_{k=1}^{j-1} X_k \right) \left(\prod_{k=1}^{j-1} X_k \right) (c_j^\dagger + c_j) X_j (c_{j+1}^\dagger + c_{j+1})$$

$$Z_j Z_{j+1} = \left(\prod_{k=1}^{j-1} (X_k)^2 \right) (c_j^\dagger + c_j) X_j (c_{j+1}^\dagger + c_{j+1})$$

$$Z_j Z_{j+1} = (c_j^\dagger + c_j) X_j (c_{j+1}^\dagger + c_{j+1})$$

$$X_j = 1 - 2c_j^\dagger c_j$$

$$Z_j Z_{j+1} = (c_j^\dagger + c_j) (1 - 2c_j^\dagger c_j) (c_{j+1}^\dagger + c_{j+1})$$

$$(c_j^\dagger + c_j) (1 - 2c_j^\dagger c_j) = c_j^\dagger (1 - 2c_j^\dagger c_j) + c_j (1 - 2c_j^\dagger c_j)$$

$$(c_j^\dagger)^2 = (c_j)^2 = 0$$

$$c_j^\dagger (1 - 2c_j^\dagger c_j) = c_j^\dagger - 2(c_j^\dagger)^2 c_j = c_j^\dagger$$

$$\begin{aligned} c_j (1 - 2c_j^\dagger c_j) &= c_j - 2c_j c_j^\dagger c_j = c_j - 2(1 - c_j^\dagger c_j) c_j \\ &= c_j - 2(c_j - c_j^\dagger (c_j)^2) = c_j - 2c_j = -c_j \end{aligned}$$

$$Z_j Z_{j+1} = (c_j^\dagger + c_j) X_j (c_{j+1}^\dagger + c_{j+1}) = (c_j^\dagger - c_j) (c_{j+1}^\dagger + c_{j+1})$$

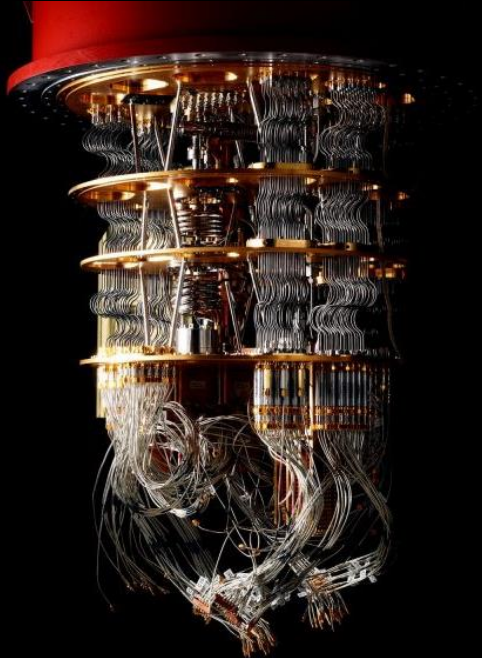
$$Z_j Z_{j+1} = c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1} - c_j c_{j+1}^\dagger$$

$$-J \sum_{j=1}^N Z_j Z_{j+1} = -J \sum_{j=1}^N c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1} - c_j c_{j+1}^\dagger$$

$$-h \sum_{j=1}^N X_j = -h \sum_{j=1}^N 1 - 2c_j^\dagger c_j = -h \sum_{j=1}^N 1 + h \sum_{j=1}^N 2c_j^\dagger c_j$$

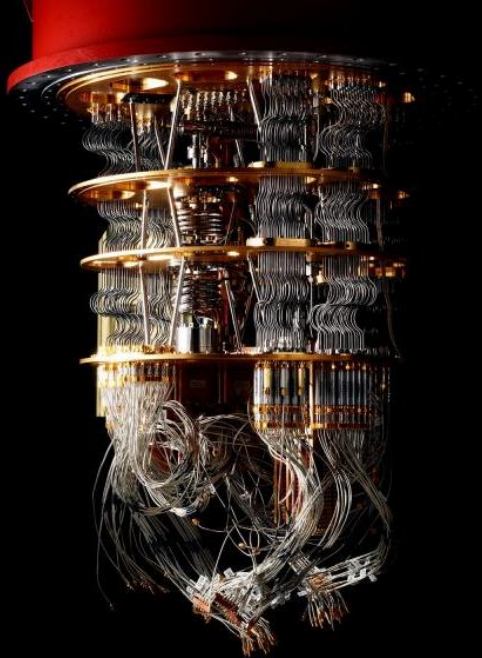
$$-h \sum_{j=1}^N X_j = -h \sum_{j=1}^N 1 - 2c_j^\dagger c_j = -hN + h \sum_{j=1}^N 2c_j^\dagger c_j$$

$$H = -J \sum_{j=1}^N (c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1} - c_j c_{j+1}^\dagger) + 2h \sum_{j=1}^N (c_j^\dagger c_j) - hN$$



Tool #2: Fourier.

$$H = \sum_k \left(2(h + J \cos k)(c_k^+ c_k - c_{-k} c_{-k}^+) + 2iJ \sin k (c_k^+ c_{-k}^+ + c_{-k} c_k) \right) - hN$$



Tool #3: Bogoliubov transformations.

$$H_k = \epsilon_k (c_k^+ c_k - c_{-k} c_{-k}^+) + \Delta_k (c_k^+ c_{-k}^+ + c_{-k} c_k)$$

$$\begin{aligned} c_k &= u_k \gamma_k + v_k \gamma_{-k}^+ & c_{-k} &= u_k \gamma_{-k} - v_k \gamma_k^+ \\ u_k^2 + v_k^2 &= 1 \end{aligned}$$

$$\begin{aligned} H_k &= \epsilon_k \Big((u_k^2 - v_k^2) (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+) + 2u_k v_k (\gamma_k^+ \gamma_{-k}^+ + \gamma_{-k} \gamma_k) \Big) \\ &\quad + \Delta_k \Big((u_k^2 - v_k^2) (\gamma_k^+ \gamma_{-k}^+ + \gamma_{-k} \gamma_k) - 2u_k v_k (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+) \Big) \end{aligned}$$

$$\begin{aligned} H_k &= \Big(\epsilon_k (u_k^2 - v_k^2) - 2\Delta_k u_k v_k \Big) (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+) \\ &\quad + \Big(2\epsilon_k u_k v_k + \Delta_k (u_k^2 - v_k^2) \Big) (\gamma_k^+ \gamma_{-k}^+ + \gamma_{-k} \gamma_k) \end{aligned}$$

$$2\epsilon_k u_k v_k + \Delta_k (u_k^2 - v_k^2) = 0$$

$$H_k = \Big(\epsilon_k (u_k^2 - v_k^2) - 2\Delta_k u_k v_k \Big) (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+)$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \qquad v_k = \sin\left(\frac{\theta_k}{2}\right)$$

$$\begin{aligned} u_k^2 - v_k^2 &= \cos(\theta_k) \\ 2u_k v_k &= \sin(\theta_k) \end{aligned}$$

$$\epsilon_k (u_k^2 - v_k^2) - 2\Delta_k u_k v_k = \epsilon_k \cos(\theta_k) - \Delta_k \sin(\theta_k)$$

$$\epsilon_k \sin(\theta_k) + \Delta_k \cos(\theta_k) = 0$$

$$\begin{aligned} \tan(\theta_k) &= -\frac{\Delta_k}{\epsilon_k} \\ \cos(\theta_k) &= \frac{\epsilon_k}{\sqrt{\epsilon_k^2 + \Delta_k^2}} \\ \sin(\theta_k) &= \frac{-\Delta_k}{\sqrt{\epsilon_k^2 + \Delta_k^2}} \end{aligned}$$

$$H_k = \sqrt{\epsilon_k^2 + \Delta_k^2} (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+) = E_k (\gamma_k^+ \gamma_k - \gamma_{-k} \gamma_{-k}^+)$$